

An obstruction to a reflection-paired five-row local model

Research proof candidate; independent review requested

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For a vector on five ordered rows write $\mathbb{E}$ for its coordinate average, and let row reflection reverse the five coordinates. An even vector is fixed by reflection; an odd vector changes sign.

Proposition 1. *There is no rational local comparison model with centered columns*

$$a = z + u, \quad b = z - u, \quad c = t + v, \quad d = t - v,$$

where z, t are odd and u, v are even, satisfying the uniform interval square-free moments through degree four:

$$\mathbb{E} \prod_{j \in S} y_j = \begin{cases} 0 & |S| \text{ odd,} \\ 1/(|S| + 1) & |S| \text{ even.} \end{cases}$$

More generally, over the reals every such model has $u = 0$ or $v = 0$. No monotonicity or interval constraint is needed for these assertions.

Proof. The first moments imply $\mathbb{E}u = \mathbb{E}v = 0$. Since odd and even vectors are orthogonal, the four mixed pair moments imply $\mathbb{E}zt = 1/3$ and $\mathbb{E}uv = 0$. Suppose first that u, v are nonzero. The space of even mean-zero vectors on five rows has dimension two, so these orthogonal nonzero vectors form a basis of that space.

Set $A = ab = z^2 - u^2$ and $B = cd = t^2 - v^2$. Both are even, $\mathbb{E}A = \mathbb{E}B = 1/3$, and the triple moments give $\mathbb{E}Av = \mathbb{E}Bu = 0$ (their products with odd vectors have zero mean automatically). Thus for some real α, β ,

$$A = \frac{1}{3} + \alpha u, \quad B = \frac{1}{3} + \beta v.$$

It follows that $\mathbb{E}AB = 1/9$, contradicting the required fourth moment $1/5$.

If $u = 0$, then $a = b = z$ is an odd rational vector with $\mathbb{E}z^2 = 1/3$. Write $z = (r, s, 0, -s, -r)$; this would require $r^2 + s^2 = 5/6$. The 3-adic valuation of a nonzero sum of two rational squares is even: after removing a common power of three, the sum of the two squared 3-adic integers is a unit. But $v_3(5/6) = -1$ is odd. This is impossible. The case $v = 0$ is identical. $\square$